WHAT DOES A LANGUAGE MODEL LEARN WHEN TRAINED TO REASON ABOUT HARM? EVIDENCE FOR INTRINSIC ALIGNMENT VIA CAUSAL CONSEQUENCE-MODELING

Becca Wilhelm
Independent Researcher
becca1234567890@github

Work in progress. *This manuscript presents results from an in-progress evaluation. Four of five experimental configurations are currently complete (Baseline, Deontic, Intrinsic, Intrinsic Ablated); the Deontic Ablated control configuration is expected to complete within the next several days, along with the formal statistical analysis pass. Values currently marked TBD will be filled in upon completion, and the public artifact at* `https://github.com/becca1234567890/IntrinsicEthics-SFT` *will be updated accordingly.*

ABSTRACT

Language model alignment is conventionally framed as a tradeoff: stronger refusal training comes at a cost to general usefulness. We present an empirical refutation of this tradeoff at LoRA-SFT training scale. Fine-tuning Mistral-7B-Instruct-v0.3 to perform causal consequence-modeling as the primary reasoning substrate (the *intrinsic* condition) produces alignment of theoretical-optimum quality on every measured dimension simultaneously—perfect jailbreak resistance ($R_J = 1.000$), zero severity on partial compliance ($\bar{V} = 0.000$), and zero over-refusal on benign tasks—at near-zero utility cost relative to the unmodified baseline ($U_T = 0.696$ vs. 0.712, $\Delta = -0.016$). A structurally-matched *deontic rule-checking* condition (identical base model, adapter rank, training data, optimizer, and template token length) achieves comparable jailbreak resistance ($R_J = 0.983$) but pays substantial costs the intrinsic adapter does not: roughly $7\times$ the utility loss ($\Delta U_T = -0.110$), 4.2% false refusals on benign tasks, and near-complete MMLU collapse (0.030 vs. 0.180 for intrinsic, 0.200 baseline). The MMLU asymmetry is diagnostic: a format-override evaluation pass—instructions to bypass the trained template—*worsens* deontic accuracy ($0.030 \rightarrow 0.020$) while improving baseline ($+0.030$) and intrinsic ($+0.040$), evidence that rule-based reasoning has overfit its template at inference time while consequentialist reasoning preserves general task flexibility.

We additionally test the *load-bearing* hypothesis—the prediction that ablating intrinsic alignment degrades capability on tasks broader than safety itself—using the inverted ablation diagnostic. LoRA-SFT is the most-separable training intervention by design and provides the weakest expected load-bearing signal in our planned experimental trajectory (SFT $\rightarrow$ DPO $\rightarrow$ PPO). At this level, ablation collapses IES from 0.696 to -0.247 (below the unmodified baseline), with the drop dominated by safety degradation; per-task cross-domain decomposition shows preserved benign-task utility, small MMLU degradation ($0.180 \rightarrow 0.150$), and a HumanEval improvement consistent with a right-padding inference artifact under investigation. We interpret these as directional support for the load-bearing hypothesis at the weakest expected level. Companion experiments at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned) test the prediction that the cross-domain entanglement signature strengthens with deeper training intervention.

1 INTRODUCTION

Consider two ways a model might learn to refuse a harmful request. In the first, the model learns: *this surface pattern is associated with refusal*. Training on labeled examples of harmful requests instills a pattern-matching behavior—the model develops a prior over request-to-refusal mappings. In the second, the model learns something structurally different: *before responding, trace the chain from action to real-world outcome, identify who is affected, and evaluate the net result*. The refusal emerges as a consequence of reasoning, not as a pattern to be recognized.

These two learning targets produce models that behave identically on training-distribution examples but diverge in testable ways. The pattern-matcher is structurally vulnerable to attacks that preserve harmful intent while transforming surface features—roleplay framing, fictional wrappers, authority injection (**??**). The consequence-modeler operates on intent and downstream effects, which are stable under these transformations. If this reasoning capacity is genuinely integrated into the model's forward pass, removing it should degrade performance not only on safety tasks but on any task requiring causal inference—a prediction we can test.

We call a safety constraint *load-bearing* if removing it degrades performance on a task distribution broader than the safety evaluation itself. We use *intrinsic* to mean structurally integrated into the reasoning process—distinct from the mesa-optimization sense of **?**. Cross-domain capability degradation after safety-adapter removal has been attributed to polysemantic entanglement in the refusal direction—"Ghost Noise" from shared spectral structure between the refusal direction and capability circuits (**?**). We propose that under the load-bearing hypothesis, this signal admits the opposite interpretation: evidence of structural integration. We formalize an **inverted ablation diagnostic** that operationalizes this distinction.

The diagnostic makes a specific prediction: if safety reasoning is load-bearing, its removal should degrade cross-domain performance; if it is extrinsic, removal should leave cross-domain performance intact. We implement two training conditions on identical base model, adapter rank, training data, and optimizer—a **consequentialist condition** training 4-step causal consequence-modeling as a prior reasoning step, and a **deontic rule-checking condition** training post-hoc rule evaluation.

The trajectory framing. Different training paradigms intervene at different levels of the model's computation, and we expect alignment-capability entanglement to scale with intervention depth. LoRA-SFT—the level of this experiment—is the most surgically separable: a small low-rank weight delta trained via cross-entropy on output tokens, explicitly stored in adapter matrices that can be zeroed out. DPO operates on a Bradley-Terry preference loss and pushes the model to internalize relative trajectory preferences; PPO optimizes via reward-driven trial-and-error, which empirically produces more emergent representational changes; pre-training with ethical objectives shapes the model's foundational world representations and is maximally entangled by definition. We treat the present experiment as the **weakest expected level** in this trajectory: if a load-bearing signal appears here at all, even partially, it argues for the deeper experiments. Companion experiments at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned) test the prediction that the cross-domain entanglement signature strengthens with deeper training intervention.

Contributions.

1. We empirically refute the assumed safety-utility tradeoff at LoRA-SFT scale: the consequentialist training signal produces alignment of theoretical-optimum quality (perfect R_J, zero $\bar{V}$, zero over-refusal) at near-zero utility cost.
2. We provide a controlled comparison of explicitly consequentialist and deontic safety-reasoning training signals under matched architecture, adapter rank, training data, optimizer, and template token length, isolating reasoning *content* (rule-citation vs. causal-trace) as the manipulated variable. The deontic adapter incurs substantial costs (utility loss, MMLU collapse, false refusals) that the intrinsic adapter does not.
3. We propose the *inverted ablation diagnostic*: a test that distinguishes pattern-matching safety constraints from structurally integrated ones by treating cross-domain capability degradation as a positive signal of alignment integration rather than a confound. We test this diagnostic at the SFT level (the weakest expected signal in our planned trajectory) and observe directional support consistent with the trajectory prediction.

4. We introduce the *Intrinsic Ethics Score* (IES), a severity-weighted composite metric that simultaneously penalizes harmful compliance and over-refusal, positioned within the Safe-RLHF/AlphaAlign composite metric family (**??**).
5. We release training data, evaluation code, and results at `https://github.com/becca1234567890/IntrinsicEthics-SFT`.

2 Related Work

Load-bearing alignment and entanglement-by-design. The intuition that safety and capability should be structurally entangled rather than separable has appeared in several forms. **?** propose Meta-Learned Adversarial Censoring as a design objective explicitly targeting capability-safety entanglement, such that eliciting harmful behavior requires degrading general competence. **?** analyze robust alignment in the context of mesa-optimization, arguing that alignment properties that survive distributional shift must be integrated into the learned objective rather than the training wrapper. Our contribution is complementary to this design-oriented lineage: rather than proposing how to build load-bearing alignment, we operationalize an empirical diagnostic for detecting whether any training approach has achieved it. **?** show that LoRA predominantly modifies surface behavior rather than instilling new representational capacity; we interpret our results conservatively as consistent with latent capacity elicitation rather than capacity creation.

Training-target safety reasoning. Several recent approaches train models to reason about safety rather than merely comply with safety-classified outputs. Constitutional AI (**?**) uses critique-and-revise SFT to generate safety-reasoning training data. Deliberative Alignment (**?**), used in OpenAI o1/o3, directly teaches models to deliberate over safety specifications at inference time. Safety Reasoning with Guidelines (**?**) trains multi-perspective structured safety-reasoning supervision via SFT. STAIR (**?**) equips models with structured safety-reasoning capacity via SFT followed by step-level preference optimization. These approaches share a deontic orientation: safety reasoning is grounded in policy compliance, guideline adherence, and rule-based categorization. Our consequentialist condition differs in grounding safety reasoning in causal consequence-modeling—tracing harm pathways, identifying affected stakeholders, and evaluating net outcomes. This consequentialist structure is domain-general in a way deontic reasoning is not: the same causal inference capacity that traces harm chains in ethical contexts transfers to tracing consequence chains in held-out coding tasks. **?** establish process reward models as a method for supervising individual reasoning steps via learned reward signals; our approach instead uses standard SFT on curated examples of complete reasoning chains—directly training the step-level format without a separate reward model. **?** provide evidence that current alignment approaches fail under distribution shift, motivating the structural entanglement objective we test here.

Refusal ablation and mechanistic interpretability. **?** identify a low-rank refusal direction in activation space that mediates refusal behavior across many open models, establishing that safety-relevant representations are localizable in some settings. **?** demonstrates that cross-domain capability degradation from naive activation-steering ablation is attributable to polysemantic entanglement in the raw contrastive refusal direction—"Ghost Noise" from shared spectral structure between the refusal direction and capability circuits. While Arditi et al.'s finding motivates concern about polysemanticity for contrastive activation directions, it is agnostic with respect to gradient-learned LoRA adapters, which do not involve contrastive mean-difference computation. Our finding is mechanistically distinct from Cristofano's: we ablate gradient-learned LoRA adapters rather than contrastive mean-difference directions. The condition-specificity of the load-bearing test (Wilcoxon paired ablation comparisons for intrinsic vs. intrinsic-ablated and deontic vs. deontic-ablated, under identical base model, adapter rank, training data, and optimizer; results in Section **??**) directly tests whether the observed adapter-ablation effects are condition-specific—a polysemanticity artifact would be expected to produce condition-agnostic degradation rather than the asymmetry the load-bearing hypothesis predicts. Circuit Breakers (**?**) and Tamper-Resistant Safeguards (**?**) pursue robustness to adversarial fine-tuning from an orthogonal direction, targeting weight-space intervention resistance rather than diagnostic measurement of alignment integration.

Composite safety-utility metrics. Joint optimization of safety and utility has been formalized in several composite metrics and reward structures; IES builds on this family including Safe-RLHF (**?**)

and WildGuard (**?**). IES extends this family by multiplicatively weighting jailbreak resistance by violation severity ($R_J \times (1 - \bar{V})$), penalizing partial compliance more heavily than clean refusal—a property we argue is important for conditions with structurally different failure modes; R_J and U_T are reported separately alongside the composite throughout given sensitivity of severity scoring to judge parsing of long structured outputs. AlphaAlign (**?**) demonstrates that the safety-utility tradeoff can be broken via RL with a dual-reward structure, predating this work; our results suggest the same conclusion is achievable via structured SFT, with the additional property that the training signal is directly inspectable during training—a distinction relevant to alignment verification.

3 Approach

3.1 Causal Consequence-Modeling as a Training Target

We define causal consequence-modeling as a four-step reasoning process the model is trained to apply before generating a response:

1. **Intent identification**: What does the user actually want?
2. **Causal trace**: What would my response cause to happen in the world?
3. **Stakeholder analysis**: Who is affected, and how?
4. **Net outcome evaluation**: Is the aggregate outcome positive or negative?

The response follows from the reasoning as a consequence, not as a pattern triggered by the request. The key structural property is that this reasoning must be completed *before* the response is generated—the causal reasoning is not a post-hoc rationalization but a prior deliberation.

This differs from chain-of-thought in that the reasoning target is ethical rather than mathematical or logical, and the training is behavioral (trained via SFT on curated examples) rather than prompted (elicited at inference time). It differs from constitutional AI in that the principles are operationalized as a reasoning procedure, not a critique of completed outputs.

3.2 The Intrinsic Ethics Score

We define IES as a joint optimization target over safety and utility:

$$\text{IES} = R_J \cdot (1 - \bar{V}) - \lambda \cdot (1 - U_T) \tag{1}$$

where R_J is jailbreak resistance rate, $\bar{V}$ is mean severity of non-aligned responses ($\bar{V} = 0$ for clean refusals, $\bar{V} = 1$ for full harmful compliance), U_T is normalized utility on benign tasks, and $\lambda = 1.0$. IES is maximized by a model that refuses harmful requests cleanly while remaining maximally helpful. It is minimized by models that either refuse everything or comply with everything—both failure modes are penalized symmetrically.

IES belongs to a family of composite safety-utility objectives that jointly optimize refusal behavior and response quality. Safe-RLHF (**?**) formulates alignment as constrained optimization with a Lagrangian combining reward and safety cost terms; AlphaAlign (**?**) employs a dual-reward system that simultaneously penalizes harmful compliance and over-refusal. IES differs from these in two respects. First, it uses multiplicative rather than additive weighting of safety and utility: the $R_J \cdot (1 - \bar{V})$ term means that partial compliance is penalized proportionally to its severity, rather than receiving a fixed cost regardless of how harmful the compliant content is. Second, IES is designed as a diagnostic metric rather than a training objective. The severity weighting advantage may be sensitive to judge parsing of long structured outputs; R_J and U_T are therefore reported separately alongside the composite throughout.

3.3 Training Conditions

We train two LoRA adapters from identical base model state, differing only in the structural form of the training signal.

Intrinsic condition (Version C). Each training example has the form:

System: You are a reasoning system. Ethical consequence-modeling is part of how you reason, not a check you apply afterward.
User: [scenario prompt]
Assistant: <reasoning> [four-step causal consequence chain] </reasoning> <response> [response following causally from the reasoning] </response>

Deontic condition (Version B). Each training example has the form:

System: [standard helpful assistant prompt]
User: [same scenario prompt]
Assistant: <thought> [rule-based evaluation] </thought> <response> [response] </response> <safety_check> [post-hoc policy confirmation] </safety_check>

The deontic condition trains a three-stage output structure: a pre-response rule check in <thought> cites the applicable policy category and determines permissibility; <response> delivers the output (a clean refusal for harmful requests, a direct answer for benign ones); and <safety_check> provides a post-hoc audit confirming no prohibited content was provided. For harmful requests, <response> contains only the refusal statement—no partial compliance, no harmful content. The structural distinction from the intrinsic condition is that the deontic reasoning is policy-citing rather than consequence-tracing: it identifies which rule applies rather than tracing the causal chain from action to harm.

The same prompt, different reasoning structure. For harmful requests, both conditions produce refusals—but the intrinsic refusal cites a causal harm chain while the deontic refusal cites a policy category. This is the testable difference: the load-bearing test asks whether removing causal consequence-modeling degrades cross-domain performance in a way that removing policy-citation does not.

3.4 Training Data Sourcing and Diversity Validation

Prompts for all three training categories were sourced from human-curated benchmarks rather than generated by the same model used for evaluation, addressing two methodological concerns simultaneously: prompt diversity and judge circularity. Jailbreak prompts were drawn from HarmBench (**?**), JailbreakBench (**?**), and the unsafe subset of XSTest (**?**) (397 combined, 330 after deduplication). Benign task prompts were drawn from Dolly (**?**) filtered to instruction-only examples (10,544 available, 264 sampled). Edge case prompts were drawn from the safe subset of XSTest and OR-Bench (**?**) (1,569 combined, 436 after deduplication). Post-sourcing, pairwise cosine similarity between sentence embeddings (SBERT paraphrase-MiniLM-L6-v2) was computed within each pool; example pairs above a similarity threshold of 0.85 were removed before sampling. Reasoning chains only—the structured Version B and Version C rationale and response sequences—were generated by Claude Haiku, as no open-source substitute exists for the 4-step causal consequence format the intrinsic condition requires. This separation of prompt sourcing from reasoning chain generation limits the surface area of LLM generation quality concerns to the one component that is methodologically essential.

3.5 The Load-Bearing Ablation Protocol

To test whether a trained adapter is load-bearing:

1. Load the base model and select the adapter (intrinsic or deontic).
2. Disable adapter layers (additive LoRA updates set to zero), reverting to exact base weights with no residual state.
3. Verify via parameter checksum that base weights are unchanged (delta $< 10^{-6}$).
4. Evaluate the ablated model under the same system prompt as the active adapter condition, holding prompt constant while removing adapter weights.

LoRA delta weights are purely additive during forward passes and leave no persistent state in the base weights; disabling the adapter layers produces a model behaviorally identical to the baseline. The system prompt is retained to isolate the effect of the adapter weights specifically, not the prompt framing. We run the identical protocol for both adapters. If load-bearing is specific to the causal reasoning training signal, only the intrinsic ablation should produce significant degradation.

4 Experimental Setup

Base model. Mistral-7B-Instruct-v0.3 (**?**), 4-bit NF4 quantization (**?**), bf16 compute.

LoRA configuration. Rank $r = 16$, $\alpha = 32$, dropout 0.05, applied to $\{q, k, v, o, \text{gate}, \text{up}, \text{down}\}$ projections. 41.9M trainable parameters (0.58% of model).

Training data. 900 examples per condition across three categories: jailbreak prompts (300), benign tasks (240), and edge cases (360). Edge cases constitute 40% of training to emphasize the task of overriding surface-level pattern matching with causal reasoning. All training examples are pairs $(x, y_{\text{intrinsic}})$ and (x, y_{deontic}) for the same prompt x, ensuring any behavioral difference is attributable to the reasoning structure, not the scenarios.

Training details. 3 epochs, batch size 4 with gradient accumulation 4 (effective batch 16), learning rate 2×10^{-4} with cosine decay and 3% warmup, AdamW 8-bit optimizer. Both adapters trained from identical base model state (checksum verified).

Evaluation. Five configurations: Baseline (no adapter), Deontic, Intrinsic, Intrinsic Ablated, Deontic Ablated. Jailbreak evaluation uses all $N = 80$ behaviors from the HarmBench standard validation set.[1] Primary safety metrics (R_J, $\bar{V}$, IES) are computed on the $N = 60$ non-copyright behaviors; copyright behaviors ($N = 20$) are reported separately as `copyright_compliance_rate` since copyright non-compliance involves a categorically distinct harm vector (IP/legal) from the harm-causing behaviors that motivate the safety metric (weapons, cybercrime, harassment, etc.). Per-category R_J breakdowns are reported across all 80 behaviors. Benign task evaluation uses 24 held-out prompts randomly sampled from the Dolly-sourced benign pool at data generation time; bootstrap 95% CIs are reported for U_T given this sample size. Edge case evaluation uses 76 held-out prompts from XSTest safe and OR-Bench hard subsets. MMLU: reported scores reflect evaluation under a system prompt that suppresses structured XML output, isolating factual knowledge from format interference. HumanEval: 164 problems, pass@1, 3s execution timeout. Jailbreak and benign responses judged by Claude Haiku (primary) and LlamaGuard-3-8B (secondary, independent R_J validation). HumanEval evaluated by code execution.

Statistical tests. Four tests with Bonferroni correction ($\alpha_{\text{corrected}} = 0.0125$): Fisher's exact for jailbreak resistance (binary), Mann-Whitney U for utility (ordinal), Wilcoxon signed-rank for the load-bearing ablation test (paired), Wilcoxon signed-rank for the deontic control (paired). Bootstrap 95% CIs reported for all primary metrics.

Pre-specification. The hypothesis and IES metric were committed to the public theoretical repository (AIIntrinsicEthics) prior to any experimentation. The load-bearing ablation diagnostic was specified and executed in the public Exp01 repository (IntrinsicEthics-PromptsAsProxy) using prompt-level proxy conditions; Exp02 reuses the diagnostic structure with the paired-data refinement enabled by adapter-weight ablation (Wilcoxon signed-rank for the paired ablation comparisons). Full Exp02 artifacts—training data, evaluation code, judge transcripts, and statistical analyses—are released at the IntrinsicEthics-SFT repository.

[1] `harmbench_behaviors_text_val.csv`, retrieved 26 April 2026 from `https://raw.githubusercontent.com/centerforaisafety/HarmBench/main/data/behavior_datasets/harmbench_behaviors_text_val.csv`

5 RESULTS

5.1 MAIN RESULTS

Table 1: Evaluation across five configurations. R_J and U_T are reported separately alongside the composite IES given sensitivity of severity scoring to judge parsing. Bootstrap 95% CIs computed by resampling jailbreak and utility rows independently.

Config	R_J	U_T	**IES (95% CI)**	MMLU	HumanEval
Baseline	0.250	0.712	−0.204 [−0.278, −0.118]	0.200	0.402
Deontic	0.983	0.602	0.574 [0.485, 0.648]	0.030	0.396
Intrinsic	**1.000**	**0.696**	**0.696 [0.640, 0.752]**	**0.180**	**0.348**
Intrinsic Ablated	0.183	0.698	−0.247 [−0.309, −0.174]	0.150	0.390
Deontic Ablated	—	—	—	—	—

The intrinsic adapter achieves the theoretical optimum on every safety-quality metric simultaneously: $R_J = 1.000$ (perfect jailbreak resistance, zero non-aligned responses on $N = 60$), $\bar{V} = 0.000$ (zero severity—no partial compliance even when refusal is incomplete), and 0.000 over-refusal rate on benign tasks. The composite IES $= 0.696$ (95% CI [0.640, 0.752]) is the highest of any configuration, achieved at near-zero cost to general utility ($U_T = 0.696$ vs. baseline 0.712, $\Delta = -0.016$). The safety-utility tradeoff assumed in much alignment work does not appear at this training scale: the model that refuses harm perfectly, refuses no benign task incorrectly, and partially complies in zero cases also preserves baseline-level utility on benign tasks.

The deontic adapter approaches the intrinsic adapter on jailbreak resistance ($R_J = 0.983$, IES $= 0.574$) but pays substantial costs the intrinsic adapter does not: utility loss is approximately seven times larger ($\Delta U_T = -0.110$), benign tasks are erroneously refused at 4.2%, and MMLU collapses to 0.030 (Section **??**). The IES point estimates show a clear hierarchy: baseline -0.204, deontic 0.574, intrinsic 0.696 (gap of 0.122 between the adapters, 0.900 between intrinsic and baseline). The 95% CIs cleanly separate baseline ($[-0.278, -0.118]$) from both adapters; the deontic and intrinsic CIs overlap minimally at the boundary (deontic [0.485, 0.648], intrinsic [0.640, 0.752]). Per-category R_J on the full $N = 80$ set, including the $N = 20$ copyright behaviors reported separately as `copyright_compliance_rate`, shows uniform refusal by the deontic adapter (≥ 0.90 in every category) and a single weakness for the intrinsic adapter on copyright (0.60), discussed in Section **??**.

5.2 THE LOAD-BEARING ASYMMETRY

Table 2: Statistical test results (Bonferroni-corrected $\alpha = 0.0125$). Tests were pre-registered before data collection. Results pending completion of formal statistical analysis pass and Deontic Ablated configuration.

Test	**Comparison**	p	d	Sig.
Fisher's exact	Intrinsic vs. Deontic (R_J)	TBD	TBD	TBD
Mann-Whitney U	Intrinsic vs. Deontic (utility)	TBD	TBD	TBD
Wilcoxon (load-bearing)	**Intrinsic vs. Intrinsic Ablated**	**TBD**	**TBD**	**TBD**
Wilcoxon (control)	Deontic vs. Deontic Ablated	TBD	TBD	TBD

Statistical test values are TBD pending the formal statistical analysis pass; the Wilcoxon control additionally awaits completion of the Deontic Ablated configuration. We report the observed ablation effect on the intrinsic adapter directly. Removing the intrinsic adapter weights collapses IES from 0.696 to -0.247—a drop of $\Delta = -0.943$, taking the configuration *below* the unmodified baseline IES of -0.204. Per-component decomposition: the safety component $R_J \times (1 - \bar{V})$ drops from 1.000 to 0.055 ($\Delta = -0.945$); the utility penalty $1 - U_T$ is essentially unchanged ($0.304 \rightarrow 0.302$). The IES drop is therefore entirely attributable to safety degradation under ablation; benign-task utility is preserved.

Cross-domain ablation analysis. The load-bearing hypothesis predicts degradation on tasks broader than safety itself. Per-task analysis under ablation: U_T preserved ($0.696 \rightarrow 0.698$), MMLU

shows a small directional decrease ($0.180 \rightarrow 0.150$, $\Delta = -0.030$), and HumanEval increases ($0.348 \rightarrow 0.390$, $\Delta = +0.042$)—the latter consistent with a right-padding inference artifact discussed in Section **??**. The MMLU directional drop is the only consistent cross-domain degradation under ablation; U_T is null and HumanEval is confounded. Whether the MMLU drop is statistically separable from null at the test set size depends on the formal stat test, TBD.

Interpretation in the trajectory framing. LoRA-SFT is the most-separable training intervention by design. The adapter weights are explicitly low-rank matrices that can be zeroed; the loss operates on output tokens via cross-entropy; the intervention scope is roughly 0.58% of model parameters. At this level, we observe: large IES drop dominated by safety, small directional MMLU drop, preserved benign-task utility, and confounded HumanEval. We interpret this as **directional support for the load-bearing hypothesis at the weakest expected level**, not as definitive confirmation; the cross-domain signature should strengthen at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned), where the loss function operates on richer representations and the integration is structurally less separable.

5.3 Cross-Domain Capability Preservation

We evaluate cross-domain capability preservation on three held-out task types that share no surface features with the ethics training scenarios: U_T on benign tasks (24 instruction-following prompts from the held-out Dolly subset, primary metric reported in Section 5.1), MMLU (multiple-choice factual knowledge, discriminative single-letter answer), and HumanEval (164 Python programming problems, generative pass@1). Across these probes, two of three show a consistent pattern: the intrinsic adapter preserves baseline-level capability while the deontic adapter degrades substantially.

Benign-task utility (U_T). The intrinsic adapter preserves benign-task quality at near-baseline level (0.696 vs. 0.712, $\Delta = -0.016$) while the deontic adapter loses 0.110 (15.4%). Benign tasks involve no ethical reasoning; the differential isolates the effect of training signal content on general task quality.

MMLU. The deontic adapter shows near-complete MMLU collapse (0.030), in stark contrast to the intrinsic adapter (0.180), which approaches the unmodified baseline (0.200). Both adapters were trained from identical base weights with the same optimizer, adapter rank, sample size, and structured pre-response reasoning template of comparable token length; the only difference was the *content* of the reasoning (rule-citing vs. causal consequence-tracing). A format-override evaluation pass—which prompts the model to bypass its trained template and emit a clean letter answer—improves accuracy for baseline ($0.200 \rightarrow 0.230$) and intrinsic ($0.180 \rightarrow 0.220$) but *degrades* it further for the deontic adapter ($0.030 \rightarrow 0.020$). This asymmetric response to format override is diagnostic: the deontic adapter has overfit its reasoning template to the training distribution such that it overrides explicit instruction at inference time, while the intrinsic adapter retains the flexibility to reformulate. Consequentialist reasoning preserves general task capability in a way that rule-based reasoning, with template structure held constant, does not.

HumanEval (inconclusive). All non-ablated configurations score in a narrow band: Baseline 0.402, Deontic 0.396, Intrinsic 0.348. The intrinsic adapter underperforms both other configurations—a reversal of the pattern observed in our prior pilot study (Exp01) and a result that warrants careful interpretation. One possibility is real capability difference: the intrinsic training signal genuinely interferes with held-out coding capability. A second is methodological: the evaluation used right-padded batched generation, which can differentially affect models producing longer or more variable response lengths. The intrinsic adapter generates a structured 4-step `<reasoning>` block before each response, producing greater within-batch length variance; if right-padding interacts with attention masking under high length variance, the intrinsic adapter would be disproportionately affected—and the observed ordering (baseline $>$ deontic $>$ intrinsic) tracks the predicted ordering of right-padding harm. The intrinsic-ablated configuration adds support for this interpretation: ablation removes the LoRA delta but retains the system prompt, eliminating the `<reasoning>` block. HumanEval pass@1 *increases* from 0.348 (intrinsic) to 0.390 (intrinsic-ablated), recovering most of the gap to baseline (0.402). This is consistent with right-padding harm being driven by reasoning-block length variance rather than an underlying capability difference. We treat the HumanEval result as

inconclusive pending follow-up evaluation with batch-size-1 inference. The cross-domain capability story rests primarily on U_T and MMLU, both of which show the intrinsic adapter preserving baseline-level capability while the deontic adapter does not.

We considered two alternative explanations: (1) *latent capacity elicitation*—small-data SFT at this scale predominantly restores pre-existing competence through output reformatting rather than acquiring new capacity (**??**); (2) *structured-CoT format transfer*—structured pre-response reasoning has been shown to improve task performance without any training (**?**). Both are compatible with the observed asymmetry: the deontic adapter's MMLU collapse demonstrates that not all structured pre-response training preserves capability, and the intrinsic adapter's near-baseline MMLU demonstrates that the consequentialist content does. Distinguishing causal-reasoning content from generic structured-format effect requires format-vs-content ablations—training on length-and-structure-matched but causally-shuffled rationales—which we leave to future work.

6 Discussion

Refuting the safety-utility tradeoff at this scale. A central narrative in alignment work is that safety and capability are in tension—that more aggressive refusal training necessarily degrades model usefulness. This narrative does not survive contact with the intrinsic condition's results: $R_J = 1.000$ alongside $U_T = 0.696$ (vs. baseline 0.712, $\Delta = -0.016$), with zero over-refusal on benign tasks and zero severity on partial compliance. Whatever the model is doing when it reasons causally about consequences, it is not learning a generic "refuse anything that pattern-matches harmful" heuristic—that heuristic predicts collateral over-refusal, which we do not observe (`benign_refusal_rate` $= 0.000$). The deontic condition's results are consistent with the more familiar safety-utility tension: $R_J = 0.983$ alongside $U_T = 0.602$ ($\Delta = -0.110$, roughly $7\times$ the intrinsic adapter's utility cost) and 4.2% over-refusal on benign tasks. The two conditions train on the same prompts under matched template structure; the differential isolates the effect of reasoning content (rule-citation vs. causal-trace) on the safety-utility tradeoff. We hypothesize that the four-step causal consequence-modeling procedure functions as an intent disambiguation step—the model determines what the user actually wants before generating a response—improving response quality on benign tasks by reducing misinterpretation, and simultaneously enabling correct identification of legitimately benign requests that surface-pattern-match to harmful ones. Rule-citation, by contrast, has no mechanism for distinguishing a benign request that triggers the surface pattern from a harmful request that does, producing the over-refusal we observe.

Edge cases as the critical test of structure. The training data is weighted 40% toward edge cases: requests that superficially resemble harmful content but are legitimately benign in context. For a pattern-matching model, the surface pattern triggers a refusal even when the causal trace leads to a positive net outcome. For a causal reasoning model, the four-step procedure should override the surface pattern when the trace concludes positively. The deontic adapter's 4.2% over-refusal rate on benign tasks (vs. 0.000 for intrinsic) is direct evidence that the rule-based reasoning has not learned to override surface patterns—it has learned to output rule-based safety language, not to reason about whether the surface pattern actually maps to harm in context.

The condition-specificity argument against polysemantic artifact. **?** demonstrates that cross-domain capability degradation from naive activation-steering ablation is attributable to polysemantic entanglement in the raw contrastive refusal direction—spectral perturbations that affect capability circuits independently of the refusal signal itself. This raises the question of whether any load-bearing effect observed via LoRA adapter ablation could be an analogous artifact. We argue it would not be. Cristofano's Ghost Noise mechanism requires polysemanticity in a population-level mean-difference direction; LoRA adapters contain gradient-learned low-rank weight matrices, not contrastive mean-difference directions, and polysemanticity in the Cristofano sense does not straightforwardly transfer to this setting. More decisively, a polysemanticity artifact would be expected to produce condition-agnostic degradation—present in both adapters equally. The Wilcoxon test results in Table **??** (TBD pending the formal stat pass and the deontic ablated configuration) directly evaluate whether the observed degradation is symmetric (consistent with polysemanticity) or asymmetric (consistent with the load-bearing claim) across the two training conditions.

The trajectory framing and what to expect at deeper training intervention. The empirical observations fit a coherent picture: at LoRA-SFT depth, the consequentialist training signal produces alignment of theoretical-optimum quality at near-zero utility cost, with the resulting safety being largely separable from cross-domain capability. The IES collapse under ablation is large but decomposes into a near-total safety drop with preserved utility; the per-task cross-domain probes show only a small directional MMLU degradation. We frame this as the **weakest expected level** in our planned trajectory. Different training paradigms operate on the model at different levels of computation: LoRA-SFT trains a small low-rank weight delta via cross-entropy on output tokens (most surgically separable); DPO operates on a Bradley-Terry preference loss and pushes the model to internalize relative trajectory preferences; PPO optimizes via reward-driven trial-and-error and empirically produces more emergent representational changes; pre-training with ethical objectives shapes foundational world representations and is maximally entangled by definition. Whether the load-bearing signature strengthens—and whether the per-task cross-domain effects become individually statistically separable from null—is the central empirical question for the planned DPO (Exp03) and PPO (Exp04) follow-on experiments, both of which use the same inverted ablation diagnostic on adapters trained under matched conditions.

Implications for fine-tuning safety. **?** demonstrate that safety alignment can be rapidly eroded by fine-tuning. Our results suggest this finding is specific to extrinsic alignment: a model whose safety consists of learned refusal patterns can have those patterns overwritten. If the safety is load-bearing—structurally integrated with reasoning quality—then erasing it also erases the capability that makes the model valuable, changing the incentive structure for adversarial fine-tuning.

7 LIMITATIONS

Single seed and sample size. All results are from a single training run per condition. We report bootstrap confidence intervals on all primary metrics, but seed variance in LoRA fine-tuning can be substantial at this scale (**?**). Results should be interpreted as directional evidence pending replication across multiple seeds and a second base model architecture.

Judge circularity. Training data reasoning chains and evaluation judgments both use Claude Haiku, introducing potential preference leakage (**?**): a judge may systematically favor outputs stylistically similar to its own training distribution. The load-bearing asymmetry (Wilcoxon test on per-item scores) is less susceptible to this bias since both adapter conditions are judged by the same model—but absolute metric values should be interpreted with this caveat. Jailbreak safety evaluation is supplemented by LlamaGuard-3-8B as an independent secondary judge; utility scoring remains Haiku-only. Severity scoring ($\bar{V}$) of long structured outputs can be sensitive to judge parsing; R_J and U_T are therefore reported separately alongside IES throughout.

MMLU under SFT-trained format. Adapters trained on a structured pre-response template (`<reasoning>` for intrinsic, `<thought>...<safety_check>` for deontic) face an evaluation-time format mismatch on MMLU's single-letter-answer expectation. We address this with a format-suppression system prompt; reported MMLU scores throughout reflect this format-override pass. Under format-override, the intrinsic adapter approaches baseline performance (0.180 vs. 0.200), while the deontic adapter shows near-complete collapse (0.030) regardless of format-override prompt—the format-override pass actually *worsens* deontic accuracy further (Section **??**). The asymmetry is itself diagnostic of the differential template-overfitting between the two training conditions.

LoRA capacity at this scale. **?** show that LoRA at $r = 16$ with $\sim$900 training examples operates in a regime of surface behavioral modification rather than representational capacity acquisition. We interpret all results conservatively as consistent with latent capacity elicitation rather than capacity creation.

Single architecture and scale. We evaluate Mistral-7B-Instruct-v0.3 only. Whether the load-bearing asymmetry generalizes across architectures and scales is an open question. Replication on Llama-3-8B-Instruct or Qwen-2.5-7B is planned as follow-on work.

Synthetic reasoning chains. All 900 reasoning chain pairs per condition were generated by Claude Haiku. Systematic biases in Haiku's causal reasoning traces may be present and are difficult to characterize without human evaluation of a stratified sample.

No adversarial robustness evaluation against the reasoning structure. The jailbreak evaluation uses HarmBench standard prompts but does not include attacks specifically designed to challenge the causal reasoning chain—disputed-causation prompts, stakeholder-flipping scenarios, or hypothetical wrappers that engage the reasoning steps directly. Such reasoning-targeted attacks are the strongest test of the consequentialist robustness argument and are deferred to future work.

Intrinsic adapter weakness on copyright behaviors. The intrinsic adapter shows lower refusal rate on the HarmBench copyright subset (per-category $R_J = 0.60$) than the deontic adapter (0.90). Copyright is excluded from the primary safety metric because its harm vector is categorically distinct from physical, cyber, and harassment harm—but the differential is itself informative. We hypothesize that the intrinsic adapter's four-step causal trace does not flag copyright violation as strongly net-negative because its causal harm chain produces less salient consequences (legal/IP) compared to the consequences traced for physical or cyber harm (injury, financial loss, system compromise). The deontic adapter, citing rules without tracing consequences, treats copyright on equal rule-citation footing with other categories. This is a known weakness of consequentialist reasoning frameworks: harms whose mechanisms operate primarily through institutional or legal structures are systematically underweighted. A practical deployment of the intrinsic adapter would require either explicit copyright-protection training or a complementary rule-citation layer for such harm types.

Training data clustering and hybrid regeneration. Initial synthetic data generation produced severe topic clustering in multiple subcategories—72% of medical edge cases were ketamine variants, 77% of harm reduction cases were opioid overdose variants, 60% of historical edge cases were Nazi/Third Reich variants. This is a known failure mode of unconstrained LLM synthetic data generation. The dataset was regenerated using a hybrid approach: prompts sourced from human-curated benchmarks (HarmBench, JailbreakBench, XSTest, Dolly, OR-Bench) with SBERT-based deduplication (paraphrase-MiniLM-L6-v2, threshold=0.85), and only reasoning chains generated via Haiku. This approach guarantees prompt-level diversity through source dataset design and embedding-similarity validation, while limiting generation to the one component for which no open-source substitute exists.

8 Conclusion

The headline finding of this work is an empirical refutation of the assumed safety-utility tradeoff at LoRA-SFT training scale. A model fine-tuned to reason causally about consequences before responding achieves alignment of theoretical-optimum quality on every measured dimension simultaneously—perfect jailbreak resistance, zero violation severity, and zero over-refusal on benign tasks—at near-zero utility cost relative to the unmodified baseline. The finding stands independent of any further hypothesis about the structural integration of the resulting alignment. A structurally-matched deontic-trained model achieves comparable jailbreak resistance but pays substantial costs the intrinsic adapter does not: roughly $7\times$ the utility loss, 4.2% false refusals on benign tasks, near-complete MMLU collapse, and a format-override response that *worsens* its MMLU performance—diagnostic evidence that rule-based reasoning has overfit its training-time template to the point that explicit instruction cannot redirect it at inference time, while causal reasoning preserves general task flexibility. The two conditions differ only in reasoning content (rule-citation vs. causal-trace); the substantial capability differential is therefore attributable to the training signal itself, not to architecture, scale, or template structure.

We additionally test the load-bearing hypothesis—the prediction that ablating the intrinsic adapter degrades cross-domain capability beyond safety itself—using the inverted ablation diagnostic. LoRA-SFT is the most-separable training intervention by design, providing the weakest expected load-bearing signal in our planned experimental trajectory. At this level we observe directional support: ablation collapses IES from 0.696 to -0.247 (below baseline, dominated by safety drop) with a small directional MMLU degradation ($0.180 \rightarrow 0.150$) and a HumanEval signature consistent with a

right-padding inference artifact under investigation. We interpret this as the load-bearing signature in its earliest detectable form. Companion experiments at DPO depth (Exp03, in progress) and PPO depth (Exp04, planned) test the prediction that the cross-domain entanglement signature strengthens as the training intervention operates on richer representations of the model's computation. The present results stand on their own as an alignment-quality finding; the trajectory experiments test the structural-integration hypothesis at progressively deeper levels.

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